

The Data science process

\* Data Science is an interdisciplinary field that manages the process that can transform hypotheses and data into actionable predictions.

Definition : It is the study of data to extract meaningful insights for business. It is a multi disciplinary approach that combines principles & practices from the fields of mathematics, statistics, artificial intelligence & computer engineering to analyze large amounts of data.

⇒ Typical predictive analytics include predicting -

- who will win an election
- what products will sell together
- which loans will default.
- which advertisements will be clicked on.

→ The data scientist is responsible for acquiring the data, managing the data, choosing the modeling technique, writing the code & verifying the results.

\* Famous examples of data science

- Amazon's product recommendation systems
- Google's advertisement valuation systems
- LinkedIn's contact recommendation system.
- Twitter's trending topics.
- Walmart's consumer demand projection systems.

- The data scientist is responsible for guiding a data science project from start to finish.
- Success of a data science project can be achieved by having quantifiable goals, good methodology, cross-discipline interactions & a repeatable workflow.

### \* The Roles in a Data Science Project

1. Project Sponsor → Represents the business interests; champions the project.
2. Client → Represents end users interests; domain expert.
3. Data Scientist → Sets & executes analytic strategy; communicates with sponsor & client.
4. Data architect → Manages data & data storage; sometimes manages data collection.
5. Operations → Manages infrastructure, deploys final project results.

#### 1. Project Sponsor :-

- The most important role in a data science project is the project sponsor.
- He is the person who wants the data science result.
- They represent the business interests.
- The sponsor is responsible for deciding whether the project is a success or failure.
- The data scientist may fill the sponsor role, if they can represent the business needs.  
(but it's not optimal arrangements)

\* The Ideal sponsor meets the following condition:

- If they are satisfied with the project outcome, then the project is defined to be a success.
- ⇒ Getting sign off becomes the central organizing goal of a data science project.
- Also, it's critical to keep the sponsor informed & involved. Show them plans, progress & intermediate successes or failures.
- ⇒ To ensure sponsor sign-off, <sup>we must get</sup> clear goals from them through directed interviews.
- we should capture the sponsor's expressed goals as quantitative stmts.

## 2. client

- ~~The~~ sponsor is the role that represents the business interests.
- ||ly The client is the role that represents the model's end users interest.
- Sometimes the sponsor & client roles may be filled by the same person.
- client is more hands-on than the sponsor.
- They are the interface b/w the technical details of building a good model & the day-to-day work ~~process~~ process into which the model will be deployed.
- They are not necessarily strong mathematically or statistically but are familiar with the relevant business process into which the model will be deployed.



- They should serve as a domain expert on the team.
- In addition to that of sponsor, client also should be informed & involved.
- Regular meetings should be <sup>arranged</sup> to keep your efforts aligned with the needs of the end users.
- Generally, the client belongs to a different group in the organization & has other responsibilities beyond projects.
- Keep meetings focused ~~on~~, present results & progress in terms they can understand & also take their critiques & comments.
- \* <sup>Practically,</sup> If the end user can't or won't use the model, then the project isn't a success in the long run.

### 3. Data Scientist :

- The next role in a data science project is the data scientist, who is responsible for taking all necessary steps to make the project succeed, including setting the project strategy & keeping the client informed.
- \* They design the project steps, pick the data sources & pick the tools to be used.
- As they choose the ~~new~~ techniques that will be tried, they should have good knowledge ~~also~~ on statistics & machine learning.
- Also responsible for project planning & tracking.
- At technical level, the data scientist also looks at the data, performs statistical ~~new~~ tests & procedures, applies ML models & evaluate results. (Total science part of Data Science)

### 3. Data Architect

- \* The data architect is responsible for all the data & its storage.
- This role is filled by a person, someone outside the data science group such as DBA (Database Administrator) or architect.
- These Data architects manage ~~your~~ data warehouse for many different projects & they may <sup>be</sup> available for quick consultation.

### 4. Operations :

- \* The operations role is critical, both in acquiring data & delivering ~~the~~ the final results. usually has operational responsibilities outside the DS group.

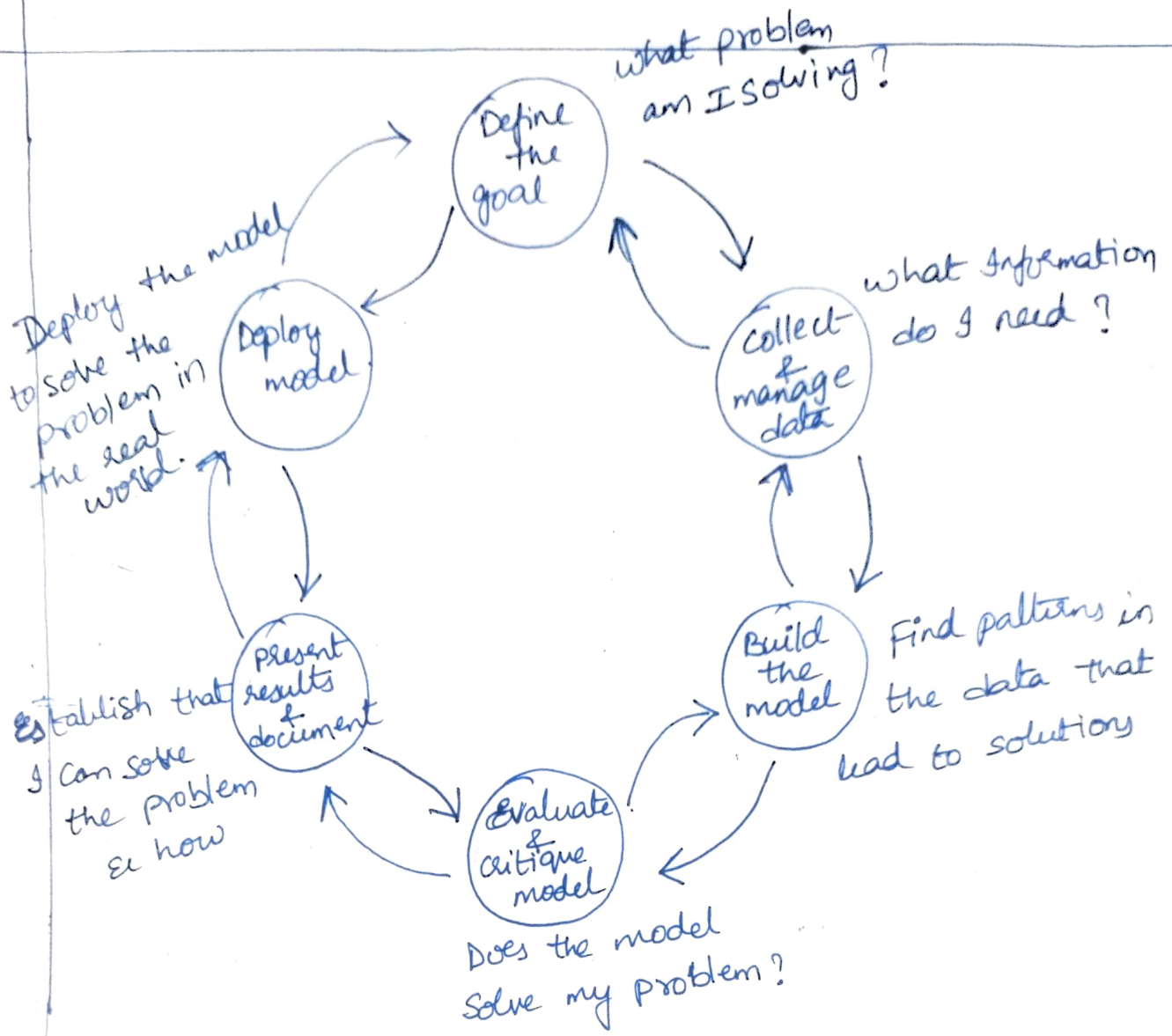
### \* Stages of a Data science project

- The Ideal data science environment is one that encourages feedback & iteration b/w the data scientist & all other stakeholders.

#### → Stages :

- Define the goal
- collect & manage data
- Build the model
- evaluate & critique model
- present results & document
- Deploy model.

3, 5, 13, 15, 27, 28, 29  
34, 36, 37, 40, 45, 56,  
1, 4,



## 1. Define the goal

- The first task in a data science project is to define a measurable & quantifiable goal.
- At this stage learn all that you can about the content of your project.
- why do the sponsors ~~also~~ want the project in the first place?
  - why do they lack, & what do they need?
  - what are they doing to solve the problem now & why isn't <sup>that</sup> good enough?
  - what resources will you need? what kind of data & how much stuff?

- will you have domain experts to collaborate with & what are the computational resources?
- how do the project <sup>sponsors</sup> plan to deploy your results?  
what are the constraints that have to be met for successful deployment?

example: Business goal: reduce the bank's losses due to bad loans.

- So project sponsor envisions a tool to help loan officers more accurately score loan applications & so reduce the number of bad loans made.
- once all the members establish preliminary response to these questions - they should define a precise goal the project ~~the~~ that is specific & measurable.  
not like: "we want to get better at finding bad loans."  
instead: "we want to reduce our rate of loan charge-off by at least 10% using a model that predicts which loan applications are likely to default."
- ⇒ The concrete goal leads to concrete stopping conditions & concrete acceptance criteria.
- ⇒ If you don't know what to achieve, you don't know when to stop trying (or even what to try).
- ⇒ once you have a good idea of the project goals, you can focus on collecting data to meet the goals.



## 2. Data collection & Management

- This step is about identifying the data you need, exploring it, & conditioning it to be suitable for Analysis.
- this stage is the most time-consuming step in the process & one of the most important as well.
  - what data is available to me?
  - will it help me solve the problem?
  - Is it enough?
  - Is the data quality good enough?
- \* This is the stage where you initially explore & visualize your data. & also clean the data: repair data errors & transform variables as needed.